

Next Gen Speech Recognition for Enterprise

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Speech recognition is in a rut. We're the physicist outsiders fixing it.

Starting with the largest dataset in the world:
Enterprise Speech Recognition.

DEEPGRAM

We sell to these leaders in Enterprise & Medical

Product Leaders

- Voice Platform
- Strong Engineering Team
- Forward Thinking

Data Science Teams

- Large Call Centers
- Established Data Science Team
- Voice Data Goldmine

Problem

Speech recognition is **not solved** for phone calls or meetings, i.e. doesn't work for enterprise.

CONSUMER MARKET

93% ACCURACY Commands

ENTERPRISE MARKET

71% ACCURACY Phone Calls

65% ACCURACY Meetings

Enterprise companies can’t adopt voice understanding at scale yet.

Solutions from Legacy Competitors don’t work. Big Tech solutions are just placeholders.

The current experience:

60-75%	???	...
ACCURACY	UNRELIABLE	MISSING FEATURES

Solution

Deepgram provides fast, reliable speech recognition that **learns from customer audio** to get the best accuracy.

Data Driven DL API

Fast, reliable, scalable, low cost

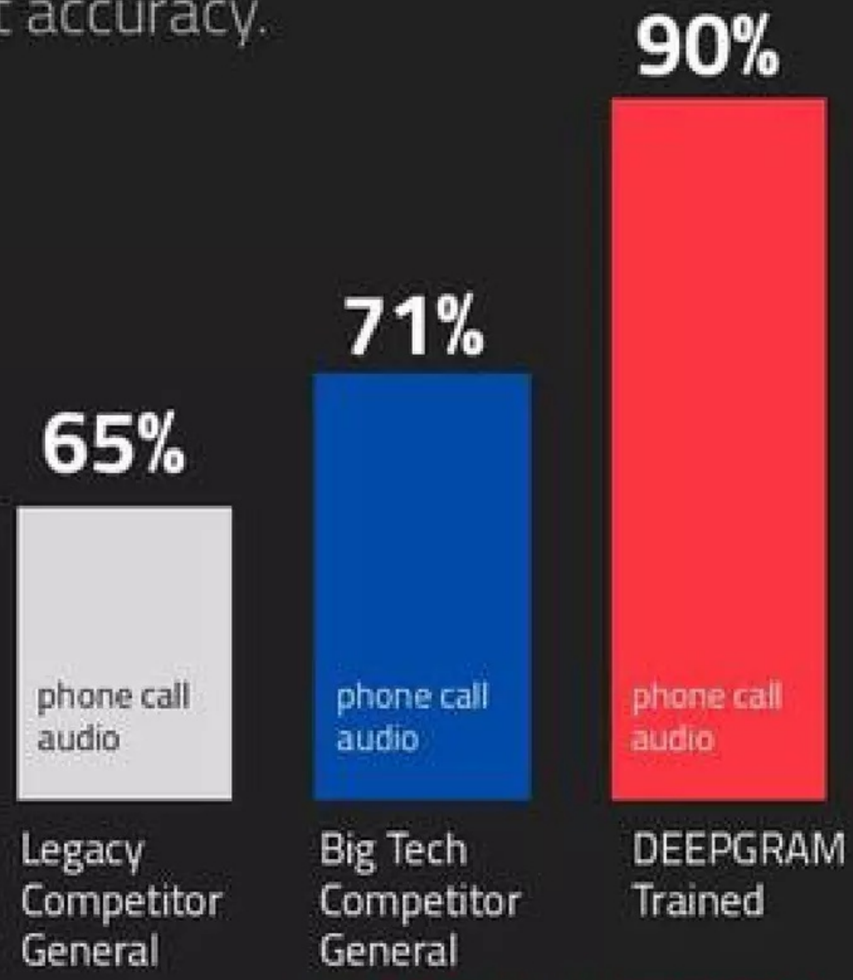
Data Flywheel

We train to get the best accuracy

raw data → label → train → deploy

Patented Architecture

Lean, low cost, truly innovative



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How do we do it?

SOLUTION

The tech is groundbreaking, unmatched, and patented.

DATA we get raw data from customers and label in house

+

MODEL we use end to end deep learning to increase speed of model building and reduce the problem to be purely data-driven

+

COMPUTE we host models and infer entirely on GPUs to provide massive throughput (which reduces system complexity and cost)

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It's universal.

Our architecture allows models to learn anything a human can do.

- Learn new languages at 10X speed, detect them too
- Figure out who is speaking
- Determine demographic information about the speaker (a/s/l?)
- Gauge their mood or sentiment
- Learn new vernacular, jargon, topics

All with the same data-driven model architecture

B2B Market

Enterprises want speech transcription.
They need accuracy and affordability at scale.

Big, It's Only Nice to Have
CONSUMER

95% ACCURACY NEEDED

NO WILLINGNESS TO PAY

This is where we focus

Big & Need It Now
ENTERPRISE
Sales & Support

85% ACCURACY NEEDED

$$ WILLINGNESS TO PAY

Big & Have Always Needed It
MEDICAL

99% ACCURACY NEEDED

$$ WILLINGNESS TO PAY

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What will win enterprise?

One-Size-Fits-All

Tries to serve all, but serves none

Stuck with **65-75% accuracy**
Customer **can't improve accuracy**

- Catch-all model stretched beyond capability
- Poor on calls and meetings
- Lacks enterprise features
- Slow: 2x speedup
- 1 hour transcript done in 1,800 seconds
- Unreliable: 30% of uploads fail

Legacy & Big Tech Competitors

VS.

Trained Expert

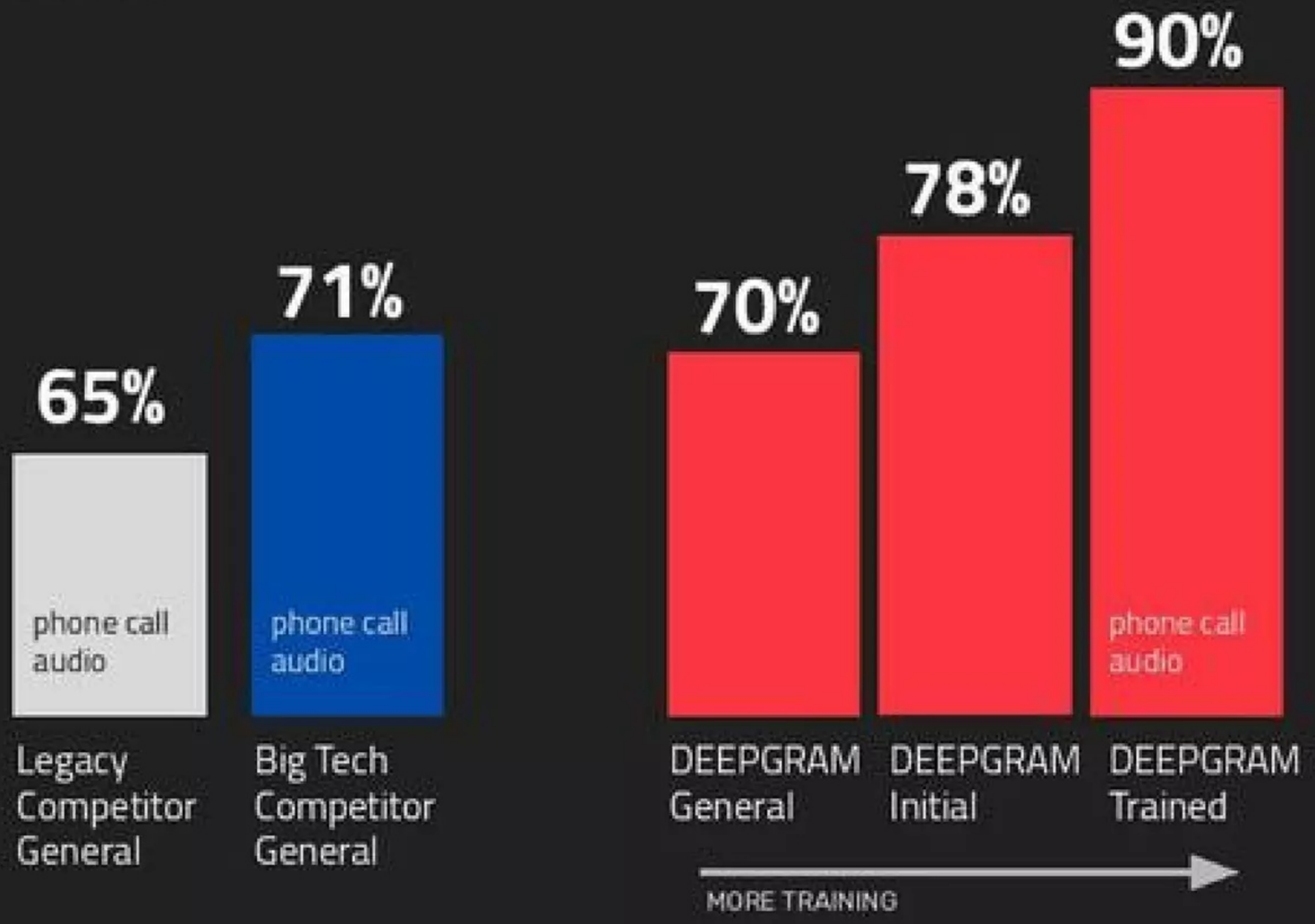
Focused on the customer

Achieves **80-95% accuracy**
Customer **CAN improve accuracy**

- Models trained for each customer
- Great on calls and meetings
- Loaded with enterprise features
- Fastest in the world: 120x speedup
- 1 hour transcript done in 30 seconds
- Rock solid availability: 99.9% SLA

DG
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Our trained model accuracy is second to none



65%

REDUCTION IN ERRORS

Just turn the dial to get more accuracy.

This is unprecedented and patented.

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B2B MARKET

FEATURES	ENABLES YOU TO	GOOGLE	AMAZON	NUANCE	IBM	DEEPGRAM
Speech-To-Text		Great	Average	Poor	Poor	Great
Realtime Streaming		Poor	None	Average	Average	Great
Expert Custom Model		None	None	None	None	Great
Deep Search		Poor	Poor	Poor	Poor	Great
Language Support		Great	Average	Average	Average	Beta
Custom Vocabulary		Poor	Poor	Poor	Poor	Great
Multi-Channel		None	None	Average	Average	Great
Timestamps		Average	Average	Average	Average	Great
Alignment		None	None	None	None	Great
Punctuation		Average	Average	Average	Average	Great
Phonemes		None	None	None	None	Great
Dedicated Cloud VPC		None	None	Poor	Poor	Great
On-Premises		None	None	Poor	None	Great
API		Average	Average	Average	Average	Great

All of these matter to Enterprise.

DEEPGRAM supports them.

DEEPGRAM

Realtime Streaming

Expert Custom Model

Deep Search

Language Support

Custom Vocabulary

Multi-Channel

Timestamps

Alignment

Punctuation

Phonemes

Dedicated Cloud VPC

On-Premises

API

Price

Understanding

B2B MARKET

THEY CARE ABOUT ACCURACY FIRST, then all of these.

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Our customers love it

TESTIMONIAL

Randall Reilly

"We tried Google's Cloud Speech API and Nuance Dragon, and investigated several other products from companies including Amazon and Prosodica.

Deepgram had the best accuracy and program by far."

90%
ACCURACY

4M
MINUTES TRANSCRIBED

Brett Evanson
CTO at Smart Rhino Labs Division at Randall-Reilly

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True AI Flywheel

We've built the right way to do transcription



Label Data
to build a unique, valuable dataset

Train Model
to get world leading accuracy

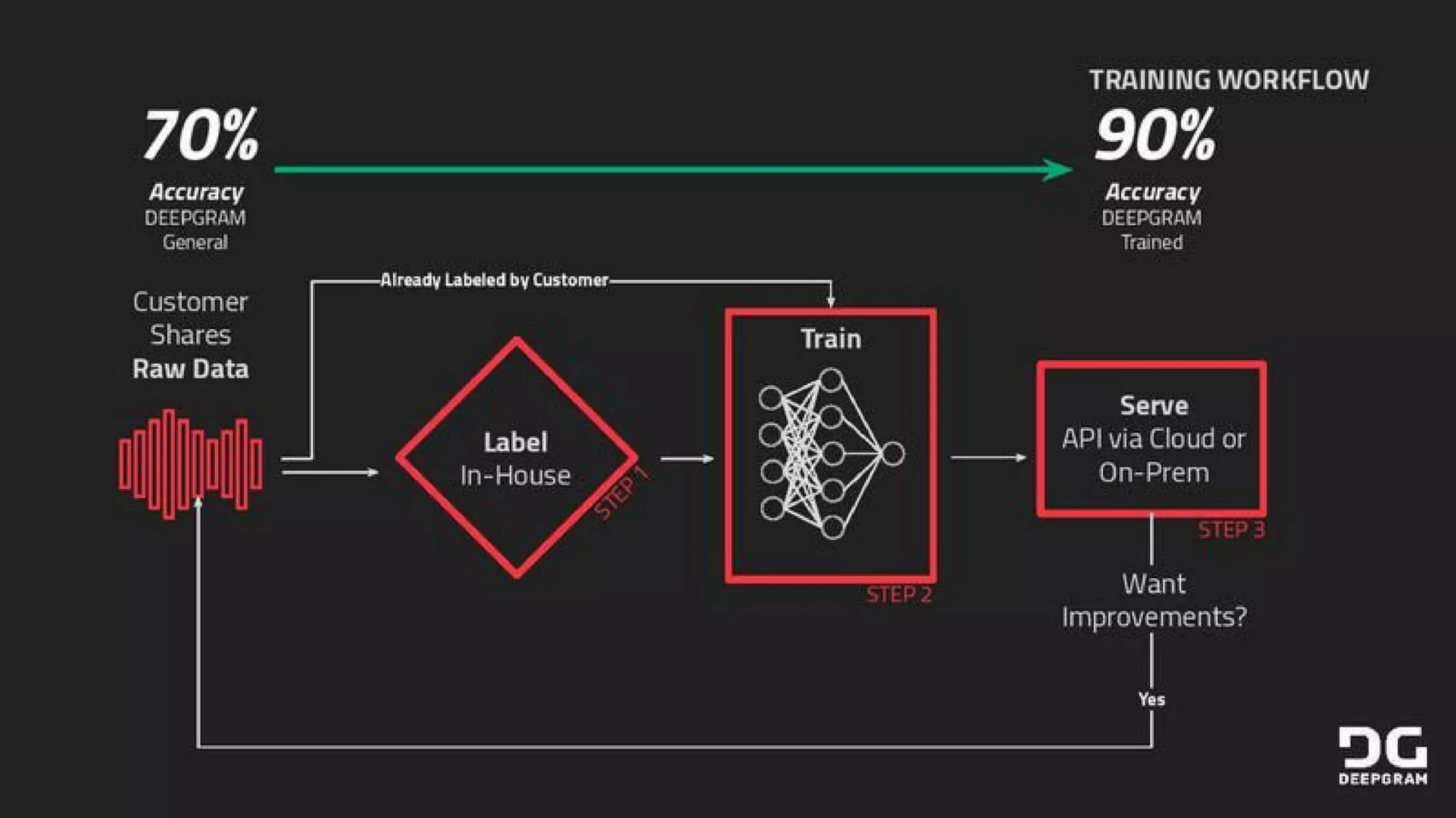
Attract More Data
with accurate, fast, affordable products

LABELING & TRAINING BOTH DONE BY DEEPGRAM IS EXTREMELY IMPORTANT.

END-TO-END DEEP LEARNING ON GPUS MAKES IT FEASIBLE AND FAST.

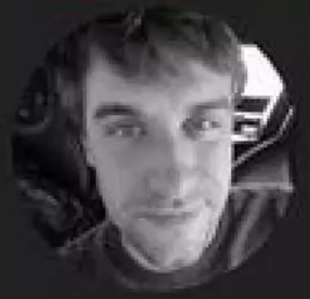
WE HAVE PATENTS FOR THIS.

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TRAINING WORKFLOW
70%
Accuracy
DEEPGRAM
General
90%
Accuracy
DEEPGRAM
Trained
Already Labeled by Customer
Customer
Shares
Raw Data
Label
In-House
STEP 1
Train
STEP 2
Serve
API via Cloud or
On-Prem
STEP 3
Want
Improvements?
Yes
DEEPGRAM

The right team for next-gen Speech AI

Leadership



Scott Stephenson
CEO

- Particle Physics PhD
- Underground Dark Matter Research
- Applied AI Systems Expert



Adam Sypniewski
CTO

- Dark Energy PhD
- Built next-gen AI tech for DARPA
- Designed AI systems for autonomous vehicles

Team

- 30+ years AI experience
- 3 Physics PhDs
- Built with 5 engineers capital efficient

DEEPGRAM

We've derisked tech, product, & sales.
Now we're going to market.

What's next:

Speech Recognition
Higher Accuracy
Languages
Lower Cost

Speech Understanding
Speaker Recognition
Sentiment Analysis
Topic Classification
Gender Detection
Age Identification
Lead Qualification
Compliance Scoring

Raise A

$3.6M Seed Raised: YCombinator, Compound, NVIDIA, Slack

Grow Team, Infra, & Data

Marketing, Sales, Engineering, Servers, Data Labeling

$10M ARR

In 18 months

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desktop → laptop → mobile → voice

Voice is the next interface

fast, natural, easy

MARKET SIZE

$12.5B

TODAY

*Grand View

We'll take the voice market by winning enterprise first.

$1T

Voice TAM

- 7B people,
- 3 mins processed by AI/day/person,
- at 1 cent/hr (200x less than GOOG today)

Even with 200x cost reduction, it's a $1.2T market

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